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Department of Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering

CSE 4142
Data Warehousing and Mining Lab

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Student Dropout and Academic Success Prediction

Lab Group: 1

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Problem Statement

Higher education institutions face big challenges in keeping students from dropping out. High dropout rates hurt both the institution's reputation and the students' future. Most current solutions act too late—only after a student has already failed. This project aims to create a smart, data-driven early warning system that can spot students at risk in their very first semester. By using information like background, family situation, and academic results, the system will predict whether a student is likely to drop out, stay enrolled, or graduate—so that timely help can be provided.

Key challenges:

- Interventions occur too late, after failure.
- Support programs are resource-inefficient.
- Insights for decision-making are unclear and non-actionable.
- Manual assessments do not scale for large student populations.

Objectives

Primary Objectives

- Develop supervised machine learning models (RandomForest, J48, SMO) to predict student outcomes in both multi-class (Dropout, Enrolled, Graduate) and binary (Dropout vs Non-Dropout) formats.
- Apply unsupervised clustering techniques (K-Means, EM) to identify natural student groupings and high-risk cohorts.
- Generate interpretable association rules that link demographic, socio-economic, and academic factors with dropout risk.
- Build production-ready, persistent models using the WEKA Java API, ensuring scalability and integration into institutional systems.

Secondary Objectives

- Perform feature engineering to derive meaningful variables (e.g., pass_rate_1st_sem, eval_load, financial_risk) that enhance model performance.
- Address class imbalance issues through SMOTE oversampling and class weighting strategies.
- Implement stratified 10-fold cross-validation for robust and reliable model evaluation with detailed performance metrics.
- Ensure interpretability by providing feature importance rankings and human-readable decision rules to guide academic policy makers and advisors.

Methodology

Dataset & Preprocessing

- Source: Higher education dataset with demographic, socio-economic, and first-semester academic data (24 attributes).
- Dataset Link: <https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention>
- Preprocessing pipeline:
 - StringToNominal for categorical features.
 - ReplaceMissingValues for missing data.
 - Standardization of numerical features.
 - Feature engineering for derived variables.

Supervised Learning

- Target: Student outcomes (Dropout, Enrolled, Graduate).
- Algorithms: RandomForest, J48, SMO.
- Validation: Stratified 10-fold cross-validation.
- Metrics: Accuracy, Precision, Recall, F1, ROC AUC, Confusion Matrix.
- Selection: Highest F1-Score for Dropout prediction.

Unsupervised Learning

- Algorithms: SimpleKMeans (k=3–6), EM.
- Evaluation: Silhouette analysis, within-cluster SSE.
- Output: Clusters profiled for dropout rates and risk identification.

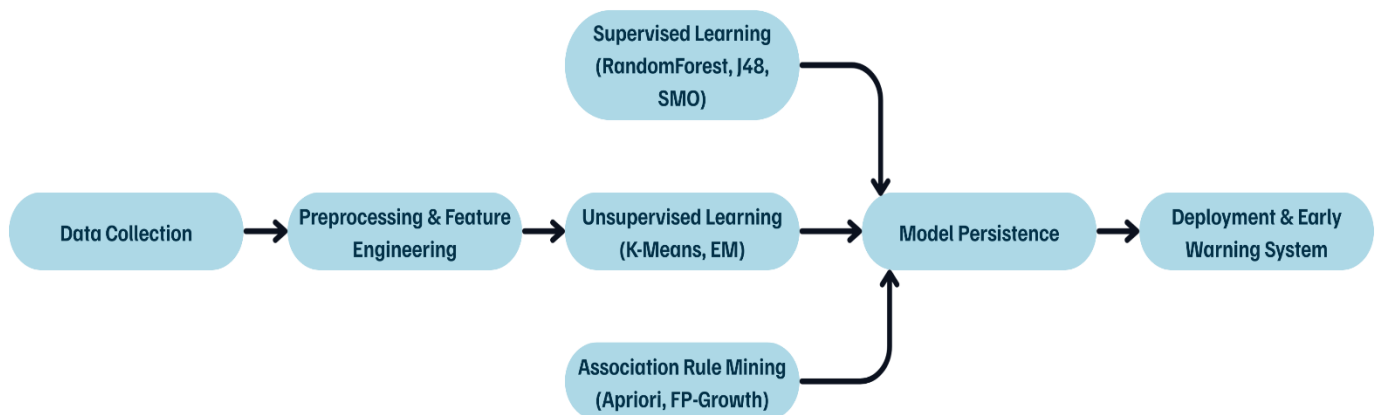
Association Rule Mining

- Preprocessing: Discretization of continuous attributes.
- Algorithms: Apriori, FP-Growth.
- Outputs: General associations and class-specific dropout rules.
- Evaluation: Rules ranked by lift and confidence.

Model Persistence

- Save optimal models using `SerializationHelper.write()`.
- Deploy in Java for real-time student risk scoring.

Workflow Diagram:



Expected Outcomes

Quantitative Targets

- Classification accuracy: >80% (multi-class), >85% (binary Dropout vs Non-Dropout).
- F1-Dropout score >0.80 with ROC AUC >0.85.
- Discovery of 4–6 clusters, including at least 2 high-risk dropout cohorts.
- Extraction of 15–25 strong association rules (confidence >0.8, lift >2.0).

Actionable Deliverables

- Early-warning system with serialized models for institutional deployment.
- Cohort profiles (e.g., financial risk, academic struggle, social integration).
- Human-readable rules and feature importance for advisors.
- Policy recommendations on intervention timing and resource allocation.

Strategic Impact

- Reduced dropout rates (estimated 15–20%).
- Optimized resource allocation to highest-risk students.
- Enhanced student success and retention.
- Scalable framework adaptable across institutions.